



Generation mechanism and emission characteristics of N₂O and NO_x in ammonia-diesel dual-fuel engine

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ABSTRACT

Ammonia, as a hydrogen-derived fuel, primarily emits N₂O and NO_x when burned in an engine. The emissions originate from ammonia fuel and nitrogen in the air. In this paper, the N of N₂ was labeled by isotope labeling method, so that the nitrogen element in the formation path of the thermal N₂O and NO_x was labeled, and the chemical reaction path of the formation and evolution of the fuel and thermal nitrogen oxide was defined. In addition, the spatiotemporal distribution characteristics, generation and evolution rules of fuel and thermal nitrogen oxide in ammonia-diesel dual-fuel engine, as well as the influence of different ammonia energy ratios on the generation ratio of two types of N₂O and NO_x were studied by combining experiment and simulation. The ammonia energy ratio is defined as the proportion of energy provided by ammonia in a constant total lower calorific value. The results showed that when the ammonia energy ratio increased from 20 % to 60 %, the emissions of NO and NO₂ decreased, but the proportion of fuel NO and NO₂ increased. The proportion of N₂O in nitrogen oxide emission increased from 7.3 % to 20 %. Fuel N₂O accounts for about 90 % of N₂O emission. Fuel N₂O and NO_x were mainly generated in ammonia-rich, low-temperature regions and thermal N₂O and NO_x were mainly generated in the high-temperature regions. Temperature significantly affected N₂O generation.

1. Introduction

Energy shortage, environmental pollution and greenhouse effects have been the main problems affecting development globally. Alternative energy currently being developed in the transportation industry to reduce carbon emissions includes electric and hydrogen energy, among others, but the vehicle power battery is currently limited by issues such as service life, endurance capacity, recharging capacity and waste post-treatment [1]. In recent years, hydrogen and ammonia have been viewed as promising zero-carbon fuels [2,3]. Although hydrogen is considered an ideal clean energy due to its high calorific value per unit mass, environmental friendliness, and recyclability [4,5], it suffers from poor safety performance and high storage and transportation requirements, which require further research. As a zero-carbon fuel, ammonia has the following traditional fuel characteristics: affordability, accessibility, storability, and transportability. Because ammonia will inevitably produce N₂O and NO_x in the combustion process, the combustion and emission characteristics of ammonia have become a research hotspot in recent years [6–12].

Otomo et al. [13] proposed an improved ammonia oxidation mechanism and simulated NH₃/air combustion in a shock tube. As equivalent ratio increases from 0.7 to 1.5, NO content decreased. Under the condition of 1.3 equivalent ratio, the content of N₂O decreases significantly. Through the contribution analysis of NO generation, it is found that NH and HNO play important roles in the process of NO generation. Philippe et al. [14] study the oxidation process of NH₃ and NO in a jet-stirred reactor (JSR). The oxidation pathway of NH₃ in JSR was given and it was found that increasing temperature resulted in increasing NO and decreasing N₂O.

Liu R et al. [15] found that ammonia has a higher octane number and using ammonia fuel can use a high compression ratio based spark ignition (SI) engines, thus improving thermal efficiency. However, because pure ammonia combustion requires very high ignition energy, compression ratio or intake temperature, its implementation in internal combustion engine (ICE) is challenging. Ammonia is therefore considered a suitable candidate to replace some carbon-based fuels (such as diesel, biodiesel, or dimethyl ether) in dual-fuel mode. Although the combustion characteristics of ammonia are not superior to those of

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methanol, ethanol or methane, its carbon-free structure, high energy density, and easy storage and transportation make ammonia an important fuel for achieving significant greenhouse gas emission reduction [16]. Notably, the data on the changes in N_2O generation caused by ammonia replacement in engines remain limited, and the N_2O contribution rate to the greenhouse effect is 273 times that of CO_2 [17]. Wolfram et al. [18] revealed that if the global shipping industry completely uses ammonia fuel for decarbonization, it may interfere with the global nitrogen cycle. Therefore, investigating the formation rules of N_2O and NO_x in ammonia fuel engines is imperative.

Merch et al. [19] studied the performance and emission of ammonia/hydrogen dual fuel in SI engines and revealed that the peak NO_x emission occurred in the range of a high hydrogen ratio and excess air coefficient of 1.1–1.4. The reason is that a large amount of hot NO_x is produced due to the high combustion temperature of hydrogen. Grannell et al. [20] and Haputhanthri et al. [21] explored the use of gasoline-ammonia dual-fuel in an SI engine. Both successfully replaced gasoline with ammonia over a limited operating range and achieved better engine performance using ammonia over a specific operating range. Westlye et al. [22] analyzed the emission level of gasoline-ammonia dual-fuel engines through tests. Due to the lower combustion temperature, NO emissions were low in mixtures near stoichiometric ratios, whereas N_2O emissions were low at optimal ignition timing and with a small amount of excess air. Ryu et al. [23] studied the effect of gaseous ammonia direct injection on SI engine performance. With gasoline providing the reference power, the total engine power increased as the ammonia injection timing was advanced, and the injection duration was lengthened. Due to the low laminar flame velocity of ammonia, the peak cylinder pressure of gasoline-ammonia dual-fuel was slightly lower than that of pure gasoline, thus reducing the overall combustion temperature. Concurrently, as the ammonia ratio increased, ammonia and NO_x emissions in the exhaust worsened.

The research of ammonia fuel in compression combustion engine is a hot topic. Reiter et al. [24] carried out a study on ammonia-diesel dual-fuel in CI engine. Through the analysis of the experimental results, the authors believe that reasonable fuel economy can be obtained when ammonia provides 40%–80 % of the total energy. Later, Reiter et al. [25] further studied the combustion and emission characteristics of ammonia-diesel hybrid fuel in a CI engine. Increasing ammonia proportion to prolong the ignition delay period was found to make most ammonia burn after the top dead center, hence decreasing peak cylinder pressure. When ammonia accounted for more than 60 % of the total energy, NO emissions increased significantly, and ammonia emissions were relatively high. Niki et al. [26] studied the effect of ammonia injection in the intake port of a single-cylinder CI engine on combustion process, aiming at reducing unburned ammonia and N_2O emissions. They found that increasing the combustion temperature prevented N_2O formation and increased the ammonia combustion rate. Notably, advanced diesel injection strategies can be used to control combustion temperature and NO_x emissions [27]. Although the preceding studies have explored the NO_x emission in ammonia engines through simulations and experiments, correlation analyses in the generation and evolution processes and emission sources of the variables remain insufficient. Lee et al. [28] studied the combustion strategy using ammonia as fuel in CI engine with a compression ratio of 35. A small amount of ammonia was pre-injected during the intake process to form a lean mixture, which naturally ignited during the compression process and ignited the main inject of ammonia. The NO formation process was divided into four stages (pre-combustion NO, reduction NO, combustion NO, and thermal NO) and the NO formation mechanism at different stages was studied. However, because different reactions occur simultaneously and affect each other in the actual process, dividing the stages of NO generation only in terms of time scale is completely inaccurate.

At present, some relevant studies on ammonia mechanism and analysis of ammonia reaction path exist and some scholars have also studied the nitrogen oxide emission characteristics of ammonia engines.

However, the combustion of mixed fuel with high ammonia ratio will emit more fuel nitrogen oxide. On the other hand, the combustion of ammonia fuel will reduce the flame temperature, thus inhibiting the emission of thermal nitrogen oxides. Therefore, a competitive relationship exists between fuel nitrogen oxides and thermal nitrogen oxides on nitrogen oxide emission content. The generation and evolution pathway of N_2O and NO_x from different sources in ammonia-diesel dual-fuel engines is currently unclear, as is the effect of varying the ammonia energy ratio on their emissions. Thus, this study explores the effects of the ammonia energy ratio on the generation and evolution of N_2O and NO_x derived from different sources by combining experiment and simulation, providing a theoretical basis for controlling their emissions from ammonia-diesel dual-fuel engines.

2. Formation and evolution mechanism and chemical reaction path of nitrogen oxides

2.1. Application of isotope labeling

The chemical reaction mechanisms used in this study were the ammonia-diesel mechanism integrated with the characterized diesel fuel (four-component mechanism comprising n-hexane, isooctane, methylcyclohexane, and toluene [29], as shown in Table 1) and ammonia mechanism [13]. Before isotope labeling, the mechanism comprised 124 species and 660 reactions. The ammonia–diesel mechanism was verified with diesel and ammonia, respectively, in terms of the ignition delay time and laminar flame speed (Figs. 1–4). Notably, the experimental data for ignition delay time and laminar flame speed were obtained from relevant studies [30–42], revealing that the ammonia–diesel combustion model accurately predicted the ignition delay time and laminar flame speed over a wide range of pressures, temperatures and equivalent ratios.

The three main sources of nitrogen oxides are thermal, prompt, and fuel sources. Because prompt nitrogen oxide is mainly produced under the condition of a highly concentrated mixture and engines rarely operate under such conditions in practice, prompt nitrogen oxide is ignored. The primary source of nitrogen oxides for diesel engines is the thermal nitrogen oxides created by N_2 in oxygen-rich air at high temperatures. For the ammonia–diesel engine, fuel N_2O and NO_x are produced due to the nitrogen contained in the fuel. Based on this, Chemkin chemical reaction kinetics software was used to label the N of N_2 in the air as N^* in order to distinguish the N_2O and NO_x after the combustion of ammonia-diesel dual-fuel. Nitrogen oxides can be divided into two sources: one is fuel N_2O and NO_x from NH_3 ; the other is thermal $\text{N}^*\text{N}_2\text{O}$ and N^*O_x from N^*N_2 .

The zero-dimensional homogeneous model was set and initial conditions of constant volume adiabatic combustion were 800 K and 10 atm. The ammonia energy ratio was set at 40 % based on a constant gross calorific value. The ammonia energy ratio is defined as the proportion of energy provided by ammonia in a constant total lower calorific value. Chemical reaction paths and sensitivity analyses of NO, NO_2 , and N_2O were performed, respectively (Figs. 5–10). The reactions involved in the formation and evolution of N_2O and NO_x were identified and labeled. Simultaneously, the analysis of the reactions with contributions to the process (Figs. 5, 7 and 9) revealed that reactions containing NH, N and HNO played important roles in the formation and evolution of N_2O and NO_x . Therefore, further reaction path tracking and reaction labeling were conducted to improve the labeling mechanism. While Table 2 lists the labeled substances, Table 3 lists part of the labeled reactions.

2.2. Validation of the isotope labeling method

The zero-dimensional homogeneous model was set with constant volume adiabatic combustion. The initial conditions were 800 K, 10 atm, and the equivalence ratio was 1. Two conditions of ammonia energy

Table 1
Parameters of the four-component mechanism.

$C_{10}H_{22}$ (mol%)	C_8H_{18} (mol%)	MCH (mol%)	C_7H_8 (mol%)	Octane number	Hydrocarbon ratio	Lower calorific value (MJ/kg)
50.39 %	10.56 %	12.86 %	26.19 %	52.17	6.83	43.43

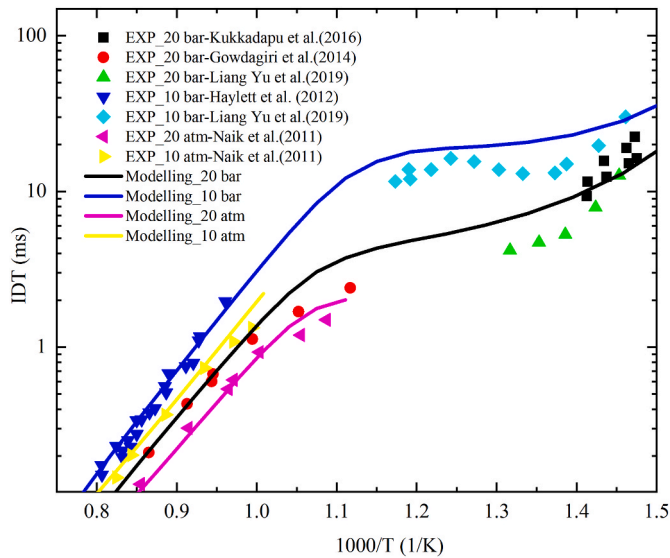


Fig. 1. Ignition delay time of ammonia–diesel model in comparison to diesel at $\phi = 1$ (Experimental values: Kukkadapu et al. [30], Gowdagiri et al. [31], Liang Yu et al. [32], Haylett et al. [33], and Naik et al. [34]).

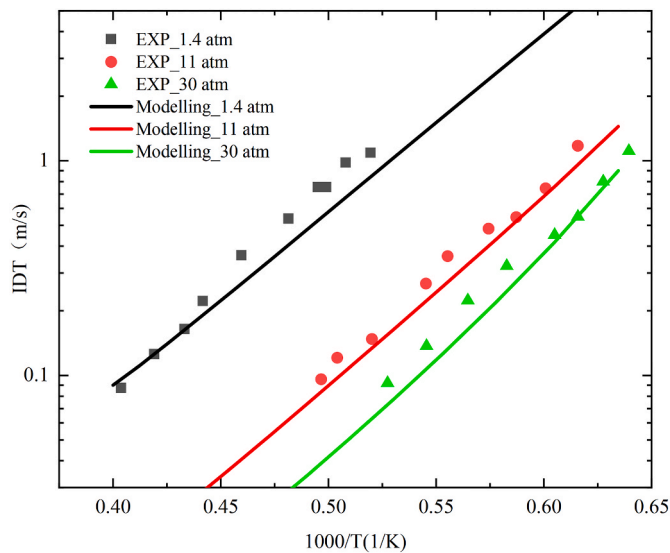


Fig. 2. Ignition delay time of ammonia–diesel model in comparison to ammonia at $\phi = 1$ (Experimental values: Mathieu and Petersen [35]).

ratios of 20 % and 40 % were set. The main substance NO in nitrogen oxides was selected as the observation object to verify the accuracy of the isotope labeling method. The NO content of different ammonia energy ratio was shown in Fig. 11 (combustion temperature, unlabeled NO content, labeled NO content, fuel NO content, and thermal N^{*}O content are shown). The content of NO was close to that before and after labeling, and the error of the content after stabilization was 3.85 % and 4.83 %.

Fig. 11 shows the accuracy of labeling under low temperature and low pressure. The accuracy of labeling under high temperature and high

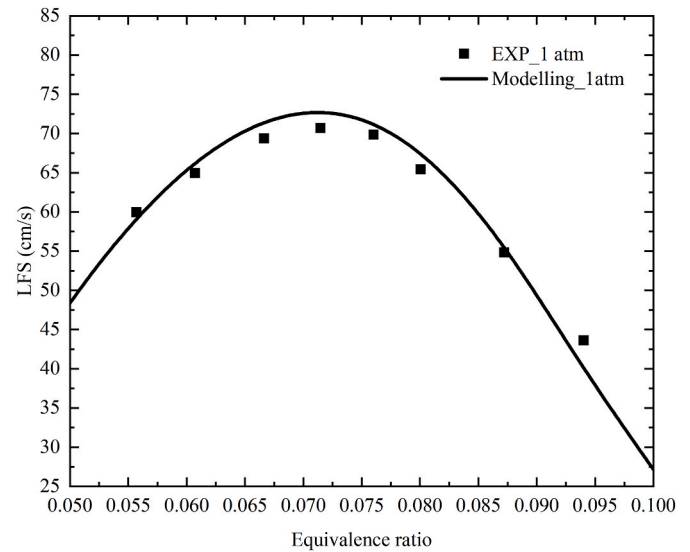


Fig. 3. Laminar flame speed of ammonia–diesel model in comparison to diesel at $p = 1 \text{ atm}$ and $T = 443 \text{ K}$ (Experimental values: B. Li et al. [36]).

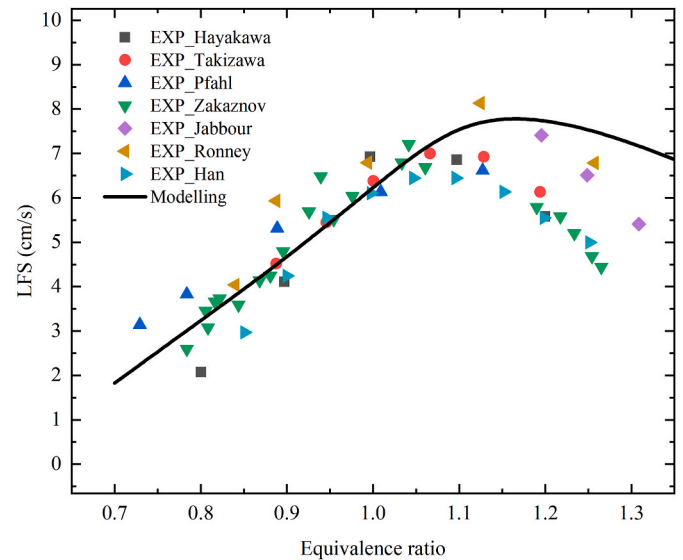


Fig. 4. Laminar flame speed of ammonia–diesel model in comparison to ammonia at $p = 0.1 \text{ MPa}$ and $T = 298 \text{ K}$ (Experimental values: 38Takizawa et al. [37], Pfahl et al. [38], Zakaznov et al. [39], Jabbour et al. [40], Ronney et al. [41], and Han et al. [42]).

pressure conditions is further verified. The constant temperature and pressure combustion method was used to revalidate the marking accuracy. In the fuel, the ammonia energy ratios was 40 %, with an equivalence ratio of 1. Four different temperature and pressure conditions—800 K–60 atm, 1000 K–60 atm, 1200 K–60 atm, and 1400 K–60 atm—were established, and the marking accuracy was verified under conditions close to the actual operating cylinder temperature of the internal combustion engine. Fig. 12 shows the results under four high temperature and pressure conditions, and the NO content before

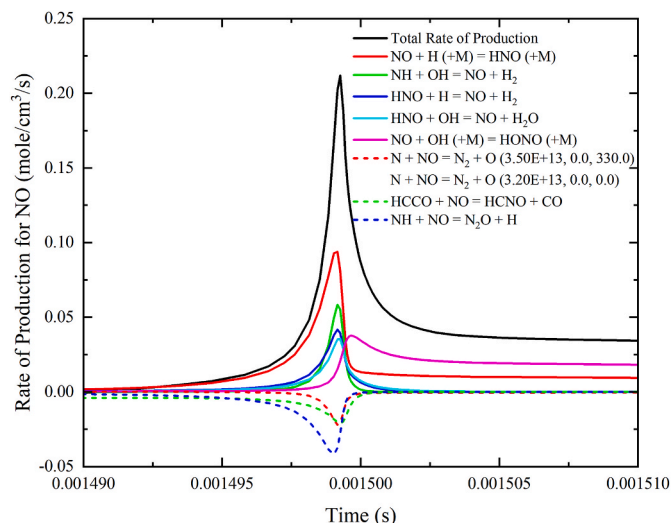


Fig. 5. Analysis of NO generation contribution.

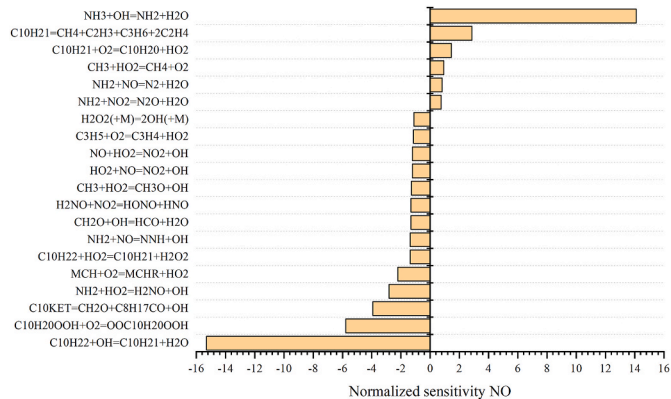


Fig. 6. Sensitivity analysis of NO.

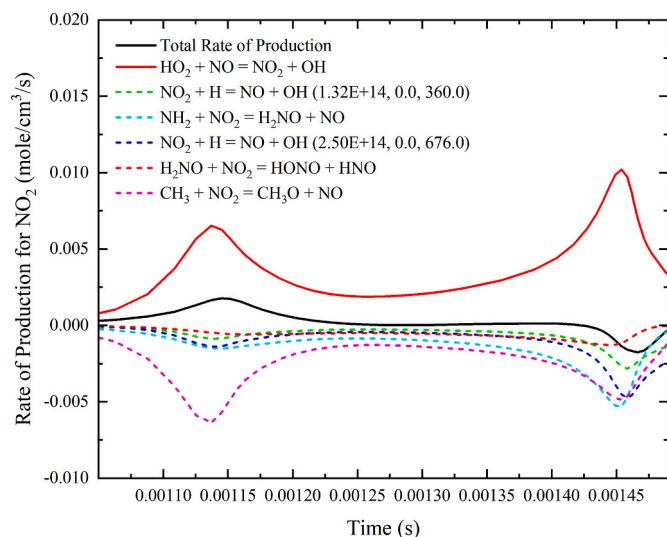


Fig. 7. Analysis of NO₂ generation contribution.

and after labeling is close. The results show that the marking accuracy is high and the mechanism can predict the emission result in the engine simulation better.

Two combustion modes of constant volume adiabatic and constant temperature and pressure were used to verify the change in NO content

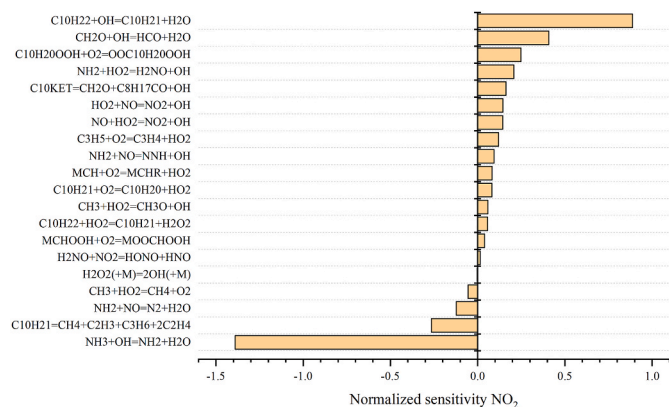


Fig. 8. Sensitivity analysis of NO₂.

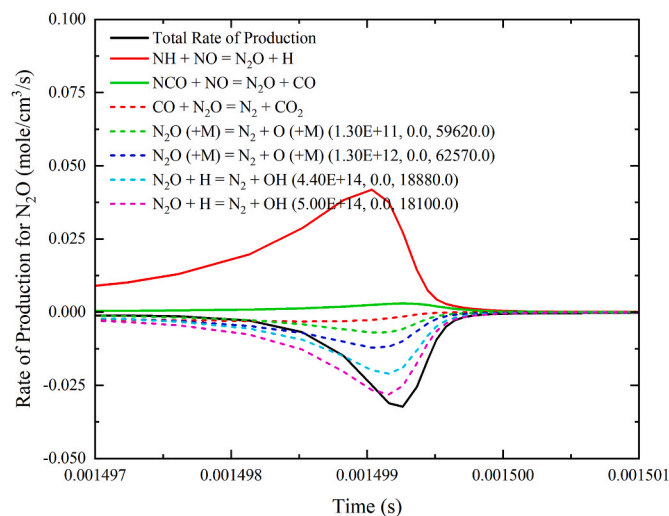


Fig. 9. Analysis of N₂O generation contribution.

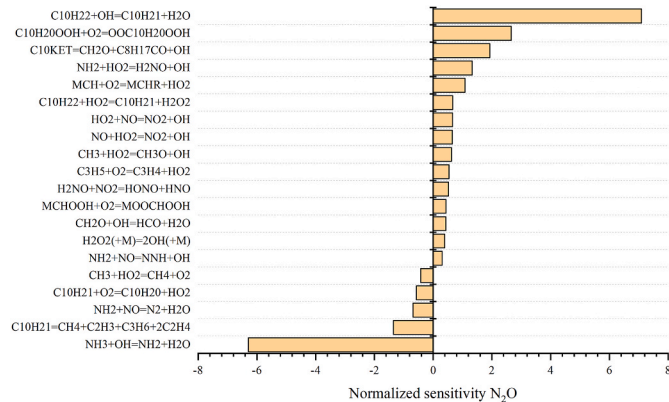


Fig. 10. Sensitivity analysis of N₂O.

Table 2

Labeled species.

Raw species	Labeled species	Raw species	Labeled species
N	N*	NH	N*H
N₂	NN*	NH₂	N*H₂
	N*₂	HNO	HN*O
NO	N*O	HONO	HON*O
NO₂	N*O₂	H₂NO	H₂N*O
N₂O	N*₂O	NNH	NN*H
	NN*O		N*N*H

Table 3
Important labeled reactions.

Reaction		Rate coefficient		
		A	n	E
R1	$N^* + N^*O = N^*_2 + O$	$3.500E + 13$	0.0	330.0
R2	$N^* + NO = NN^* + O$	$3.500E + 13$	0.0	330.0
R3	$N + N^*O = NN^* + O$	$3.500E + 13$	0.0	330.0
R26	$N^*O + O + M = N^*O_2 + M$	$1.06E + 20$	-1.4	0.0
R278	$N^*H + OH = N^*O + H_2$	$2.40E + 13$	0.0	0.0
R293	$N^*H + NO = NN^*O + H$	$2.70E + 15$	-0.8	20.0
R294	$NH + N^*O = NN^*O + H$	$2.70E + 15$	-0.8	20.0
R295	$N^*H + N^*O = N^*_2O + H$	$2.70E + 15$	-0.8	20.0
R528	$N^*O + H(+M) = HN^*O (+M)$	$1.52E + 15$	-0.4	0.0
R532	$HN^*O + H = N^*O + H_2$	$4.40E + 11$	0.7	650.0
R534	$HN^*O + OH = N^*O + H_2O$	$3.60E + 13$	0.0	0.0
R578	$N^*O + OH(+M) = HON^*O (+M)$	$1.10E + 14$	-0.3	0.0
R589	$N^*O + O(+M) = N^*O_2(+M)$	$2.95E + 14$	-0.4	0.0

before and after labeling under different ammonia replacement ratios, temperature, and pressure conditions. All simulation results showed good performance, validating the applicability of the isotope labeling method and the studied labeling method's accuracy and laying the foundation for subsequent research on the formation and evolution of nitrogen oxides from different sources.

2.3. Chemical reaction path analysis of NO

The initial conditions of constant volume adiabatic combustion were 800 K, 10 atm, and the ammonia energy ratio was 20 %. The important chemical reactions of the generation and consumption of the main substance NO in nitrogen oxides were analyzed (Fig. 13). Fuel NO was generated in large quantities in the reaction's early stages (the total NO content increasing), and its chemical reaction path is shown in the left part of Fig. 14. The main generation path was as follows: NH_3 was

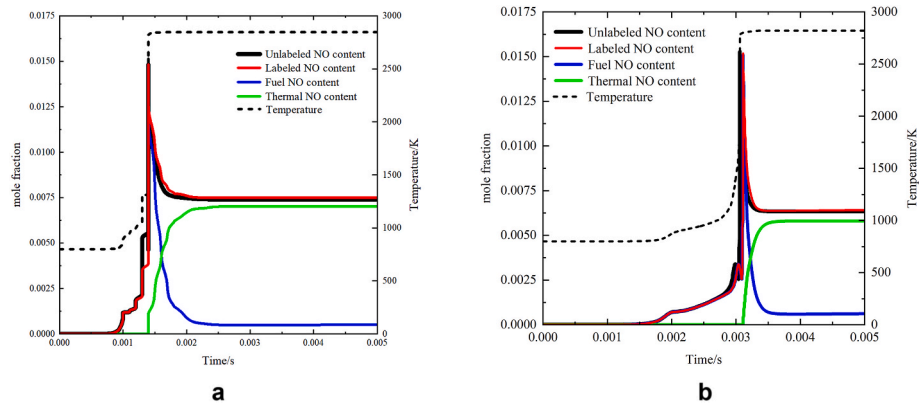


Fig. 11. NO content: (a) 20 % ammonia energy ratio, (b) 40 % ammonia energy ratio.

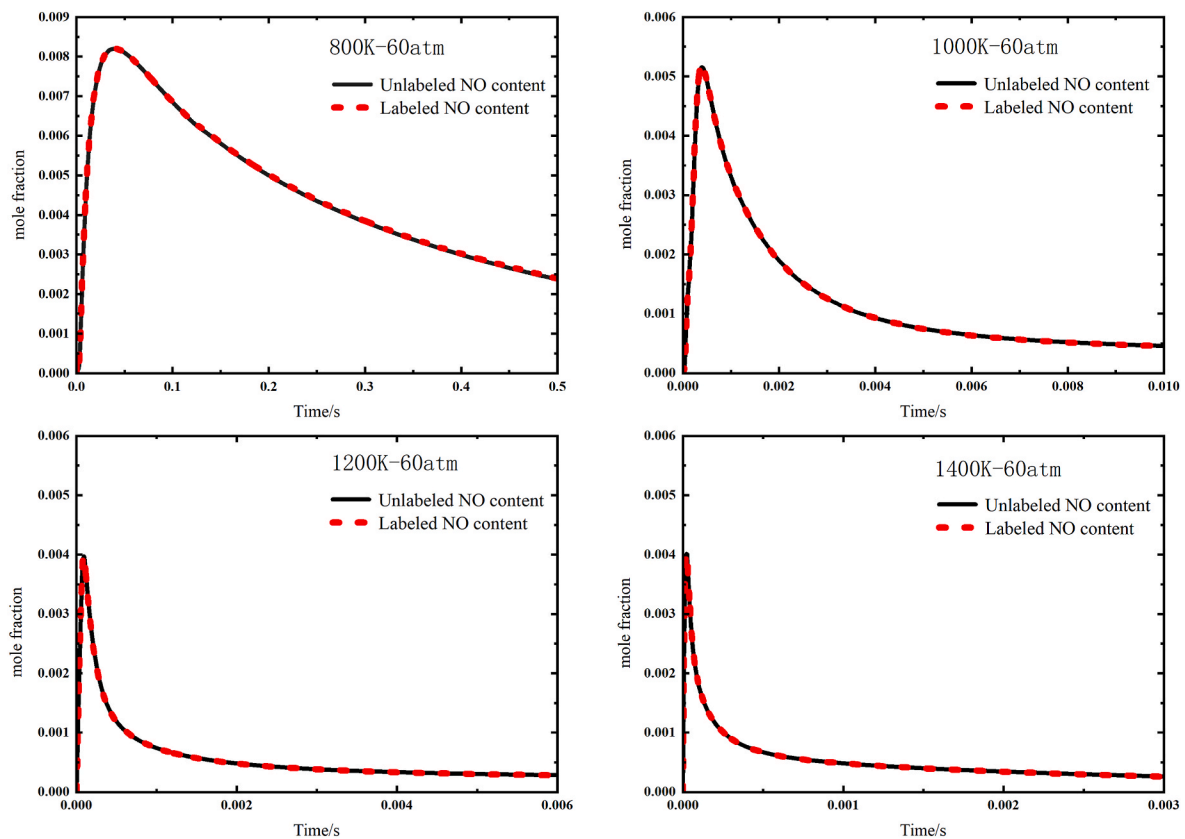


Fig. 12. NO content under constant temperature and pressure combustion conditions.

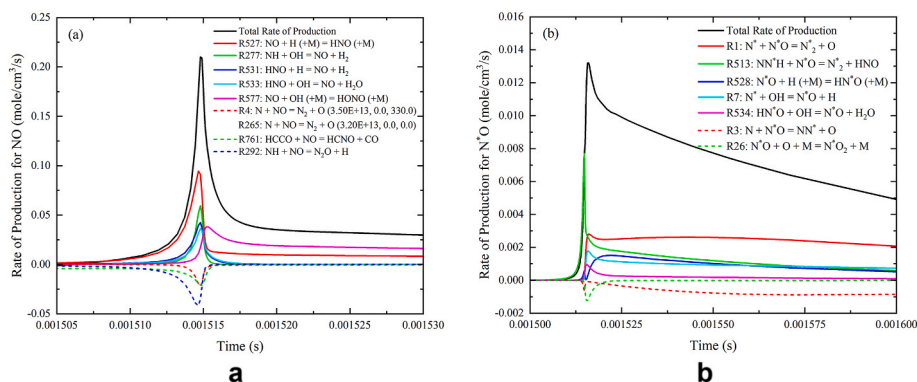


Fig. 13. Analysis of NO generation contribution: (a) fuel NO, (b) thermal N*O.

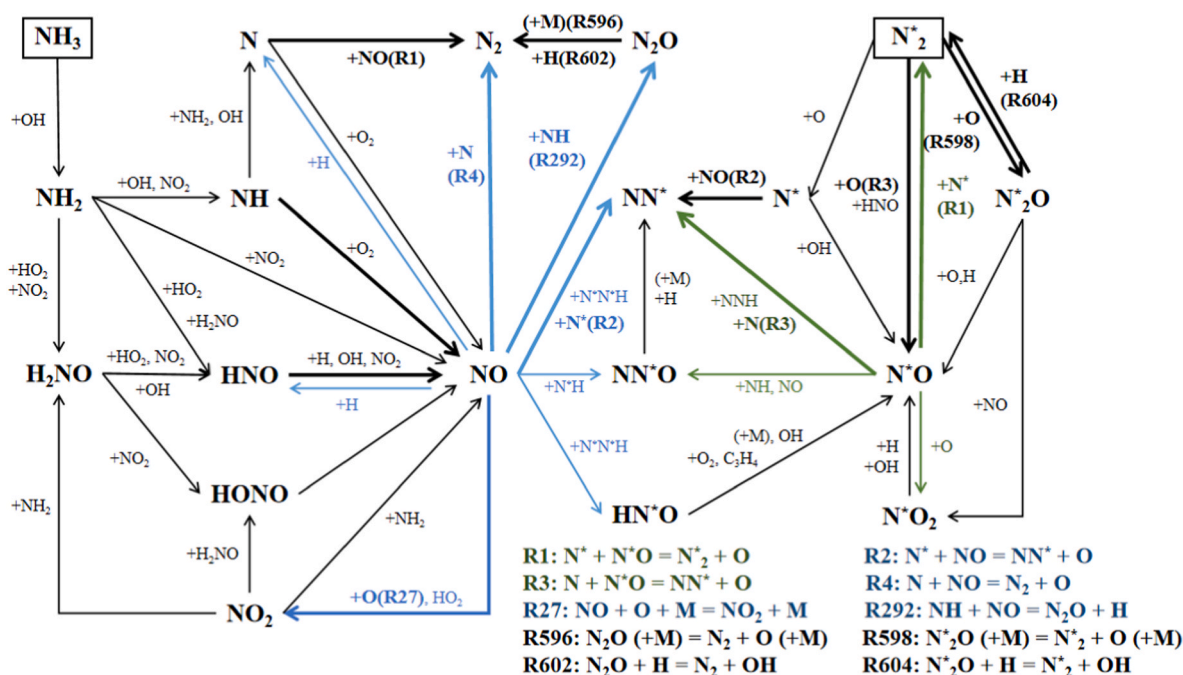


Fig. 14. Chemical reaction path of NO.

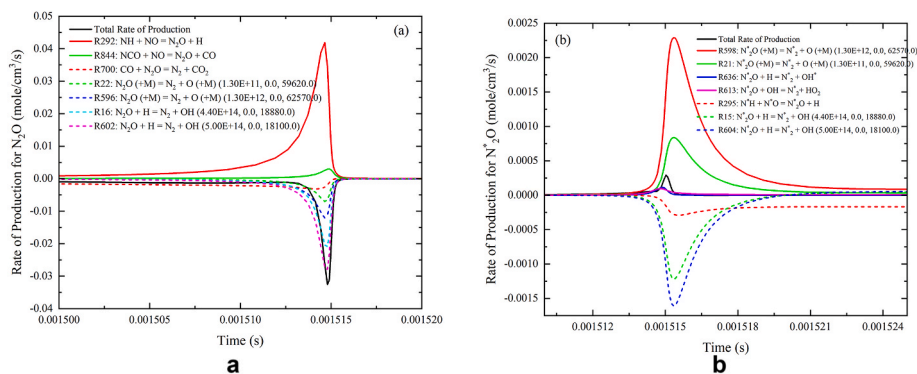


Fig. 15. Analysis of N₂O generation contribution: (a) fuel N₂O, (b) thermal N*₂O.

transformed into NH_2 , then NH and HNO , and finally fuel NO. In the reaction's late stage (the total NO content decreasing), the fuel NO content decreased significantly. The most prominent reactions that significantly affected the fuel NO content were R292: $\text{NH} + \text{NO} = \text{N}_2\text{O} + \text{H}$, R27: $\text{NO} + \text{O} + \text{M} = \text{NO}_2 + \text{M}$, and R4: $\text{N} + \text{NO} = \text{N}_2 + \text{O}$. The main

consumption path for fuel NO was oxidation to NO_2 followed by reduction to N_2O and N_2 .

Thermal N*O is hardly generated at the reaction's early stages and begins to be generated in large quantities at the late stage as the temperature rises. The thermal N*O chemical reaction path is shown in the

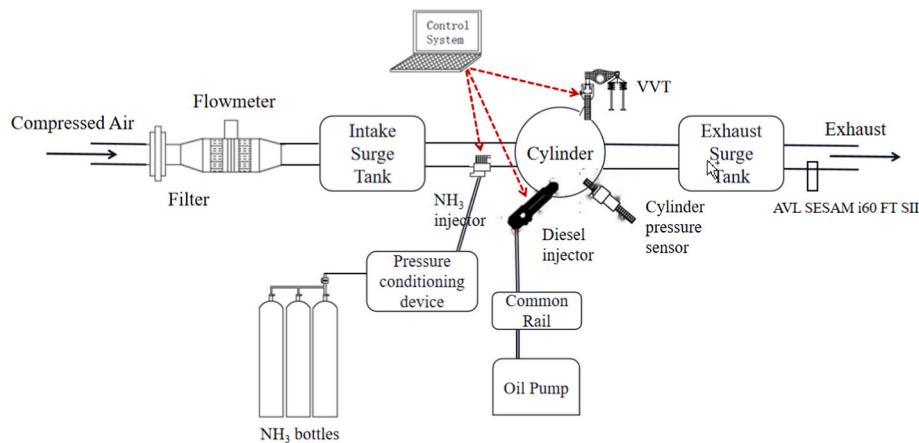


Fig. 16. The schematic of the experimental setup.

Table 4
Relevant parameters of the engine.

Cylinder diameter (mm)	Piston stroke (mm)	Connecting rod length (mm)	Compression ratio	Combustion chamber	Diesel nozzle number	Diesel nozzle diameter (mm)
116	150	245	18.5	omega-shaped	8	0.169

Table 5
The measuring accuracy of the acquisition instruments.

Measurements	Instruments	Accuracy
In-Cylinder pressure	Kstler 6125C	$\leq \pm 0.4$ bar
Air flow	TOCEIL-LFE125	± 0.5 %
Ammonia flow	DET 8622	± 0.35 %
Diesel flow	TOCEIL-CMFD010	± 0.12 %
Emissions	AVL SESAM i60 FT SII	± 1 %

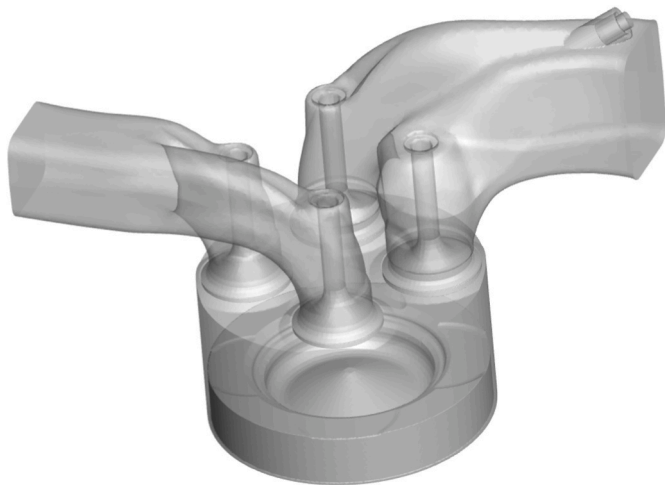


Fig. 17. CFD combustion simulation model.

right part of Fig. 14. R1: $N^* + N^*O = N^*_2 + O$ and R3: $N + N^*O = NN^* + O$ were the most prominent reactions that strikingly affected the generation and consumption of thermal N^*O . The thermal N^*O was generated by the reverse reaction of R1, and thermal N^*O was consumed by the forward reaction of R3. Under a comprehensive comparison of generation and consumption, the thermal N^*O generation rate exceeded its consumption rate, resulting in a constant increase in thermal N^*O

Table 6
Physical submodel of the 3D simulation model.

Classification	Model
Turbulence model	RNG k- ϵ model [43]
Spray model	Break-up model: KH-RT model [44] Evaporation model: Frossling model [45] Collision model: NTC and wall film model [46]
Combustion model	Combustion model: SAGE Chemical reaction solver [47] Wall heat transfer model: Han and Reitz model [48]

content.

2.4. Chemical reaction path analysis of N_2O

The important chemical reactions of the generation and consumption of the greenhouse gas N_2O were analyzed under the same initial conditions (Fig. 15). The most prominent reactions that markedly affected fuel N_2O content were R292: $NH + NO = N_2O + H$, R602: $N_2O + H = N_2 + OH$ and R596: $N_2O(+M) = N_2 + O(+M)$. Its main source was NO , which was reduced to N_2O by reaction R292. The main consumption pathways were R602 and R596, and fuel N_2O was reduced to N_2 by reacting with substances, such as H , through the above pathways.

The prominent reactions that significantly affected the thermal N^*_2O content were R598: $N^*_2O(+M) = N^*_2 + O(+M)$ and R604: $N^*_2O + H = N^*_2 + OH$. The thermal N^*_2O was mainly generated by N^*_2 oxidation and consumed by the reaction with H .

3. Combustion and emission characteristics of ammonia-diesel dual-fuel engines

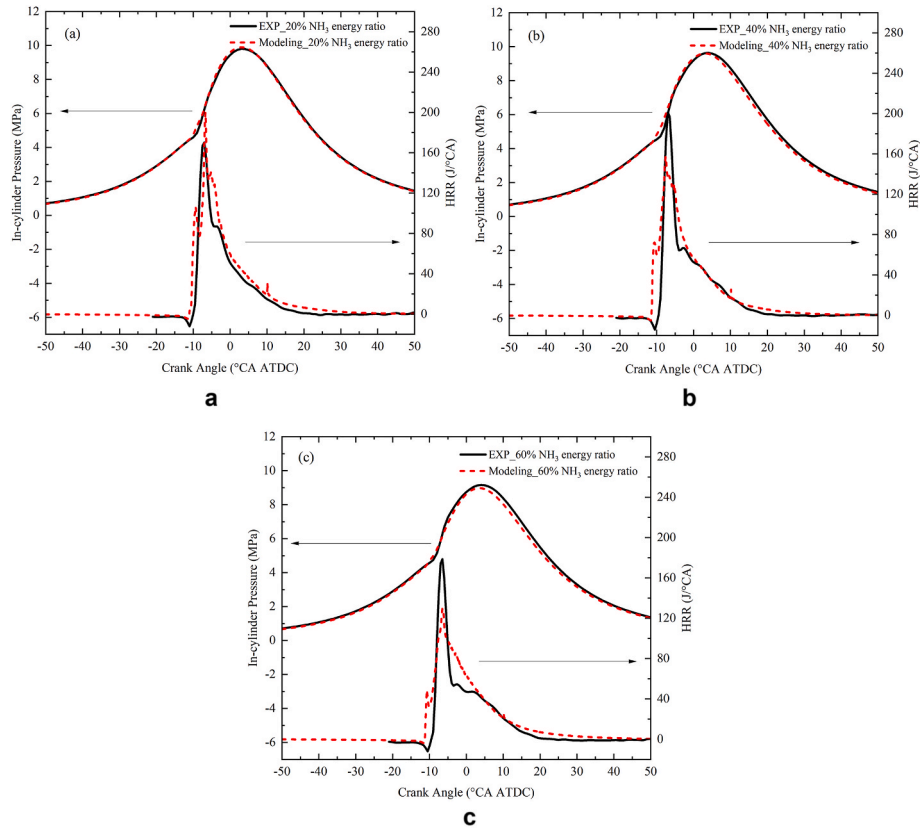
3.1. Experimental bench and model establishment

The experiment was carried out on ammonia-diesel dual-fuel engine bench. The schematic diagram and parameters of the test system are

Table 7

Calibration conditions for each ammonia energy ratio.

Speed (r/min)	Inlet pressure (bar)	Diesel injection pressure (bar)	Injection timing	Diesel quantity (mg)	NH ₃ quantity (mg)	Ammonia energy ratio	IMEP (bar)
1000	1.22	600	−14°	40	23.5	20 %	0.58
1000	1.20	600	−14°	30	47	40 %	0.58
1000	1.24	600	−14°	20	70.4	60 %	0.52

**Fig. 18.** The comparison of in-cylinder pressure and HRR between calculation and experiment: (a) 20 % NH₃ energy ratio, (b) 40 % NH₃ energy ratio, (c) 60 % NH₃ energy ratio.**Table 8**N₂O and NO_x emissions data.

NH ₃ energy ratio	NO		NO ₂		N ₂ O	
	Modeling (ppm)	EXP (ppm)	Modeling (ppm)	EXP (ppm)	Modeling (ppm)	EXP (ppm)
20 %	849.2	840.4	58.6	170.9	71.0	79.3
40 %	728.2	711.2	61.2	103.2	101.9	113.8
60 %	640.7	584.7	63.2	69.8	140.2	164.1

shown in Fig. 16 and Table 4 respectively. After passing through the pressure conditioning device, ammonia was injected into the intake manifold by the gas injection valve. It is equipped with pressure sensor to monitor ammonia supply pressure in real time. The pressure in the cylinder was measured by a pressure sensor (KISTLER 6125C), and the real-time index data was acquired by an acquisition device (NI PCIe6361). The consumption of diesel and ammonia was measured by flow meters (Toceil-CMFG010 and DET8622, respectively). Nitrogen oxide emissions were measured using a Fourier transform infrared spectrometer (AVL SESAM i60 FT SII). The measuring accuracy of the acquisition instruments used in the experiment were shown in Table 5.

The marked chemical reaction mechanism was implemented in a CFD engine combustion simulation. A CFD model was established based

on the actual engine, CONVERGE software was used to generate and divide the mesh, the mesh defects were checked, and the boundary was set. Fig. 17 shows the constructed CFD combustion simulation model, and Table 6 lists the physical submodel selected for the gas flow state and combustion process.

3.2. Experimental results and model calibration of 20 %, 40 %, and 60 % ammonia energy ratios

The calibration conditions shown in Table 7 were selected; calibration verification of the simulation model was performed using the experimental data, and N₂O and NO_x emissions were measured with AVL FTIR equipment. In the test, the total calorific value of diesel and ammonia was controlled to remain unchanged, and three ammonia energy ratios of 20 %, 40 %, and 60 % were obtained by adjusting the injection amounts of diesel and ammonia. The diesel injection pressure was 60 MPa, and that of ammonia was 6 bar.

Fig. 18 show the results of the average pressure and heat release rate in the cylinder. The peak value of the in-cylinder average pressure was about 10 MPa and decreased slightly as the ammonia energy ratio increased. The peak value of the heat release rate was higher under the condition of a 40 % ammonia energy ratio, but the combustion duration was prolonged as the ammonia energy ratio increased due to the slow ammonia combustion rate. Table 8 lists the results of N₂O and NO_x

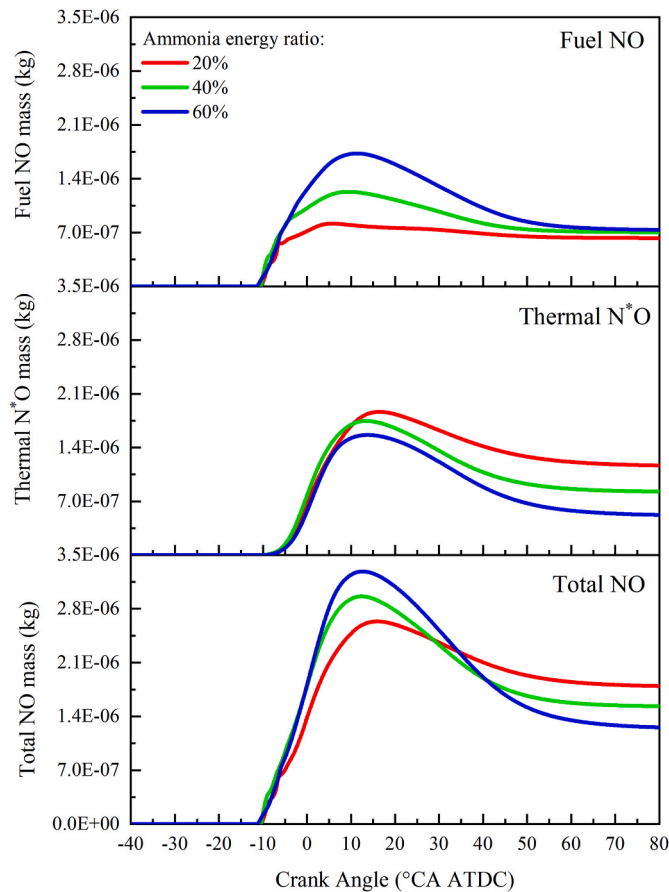


Fig. 19. Calculation results of NO content from different sources at each ammonia energy ratio.

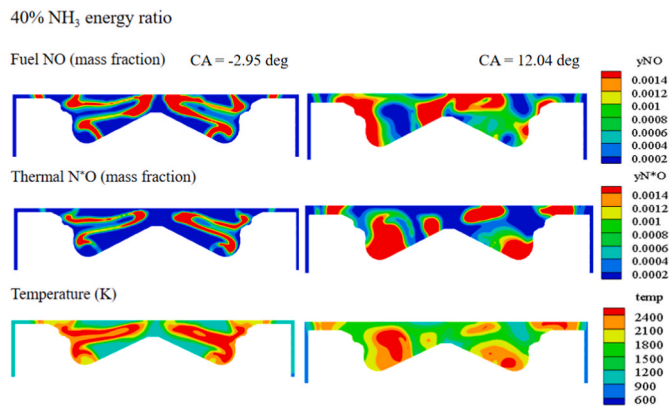


Fig. 20. Calculation results of NO and temperature distributions with a 40 % ammonia energy ratio.

emissions. When the ammonia energy ratio increased from 20 % to 60 %, NO_x emissions decreased, NO emissions were reduced by approximately 30 %, and the proportion of NO emission in nitrogen oxide emissions decreased from 77.1 % to 71.4 %. Meanwhile, N_2O emissions increased by about double, and the N_2O proportion in nitrogen oxide emissions increased from 7.3 % to 20 %. The simulation results of NO_2 are basically unchanged, which is different from the experimental results. On the one hand, the mechanism overpredicts the reaction of NO_2 , which may be the main reason for the deviation between the experiment and simulation, reflecting the shortcomings of the model. On the other hand, the accuracy of the model is also improved with the increase of ammonia energy ratio. Considering the relatively small proportion of

NO_2 in nitrogen oxides, the main nitrogen oxide component NO and greenhouse gas N_2O are mainly calibrated. It is believed that the nitrogen oxide emission of the model is similar to the actual results.

The main calibration parameters for the model were the average pressure in the cylinder, heat release rate, and N_2O and NO_x emissions. Overall, the simulated values were similar to the experimental values, confirming the established model's accuracy and laying the foundation for the analysis of nitrogen oxide emissions under different ammonia energy ratios in this study.

3.3. Analysis of NO emission characteristics

The formation and evolution of NO as a function of crank angle were studied when the ammonia energy ratio was 20 %, 40 % and 60 % (Fig. 19). For fuel NO, the maximum content increased as the ammonia energy ratio increased, but the content was almost the same after stabilization, which was caused by the consumption effect of ammonia on NO. The stable thermal N^*O content decreased as the ammonia energy ratio increased because decreased combustion temperature with an elevated ammonia energy ratio limited thermal N^*O generation, and the amino groups produced during the ammonia reaction process have a denitrification effect and can react with NO_x to finally generate N_2 . Thermal N^*O had a later generation time than fuel NO, which started after the temperature increased due to combustion in the cylinder. Notably, the total NO emission decreased as the ammonia energy ratio increased, which was attributed to the joint action of thermal N^*O and fuel NO.

For total NO, the maximum content was affected by fuel NO, and the stable content was affected by thermal NO. When the ammonia energy

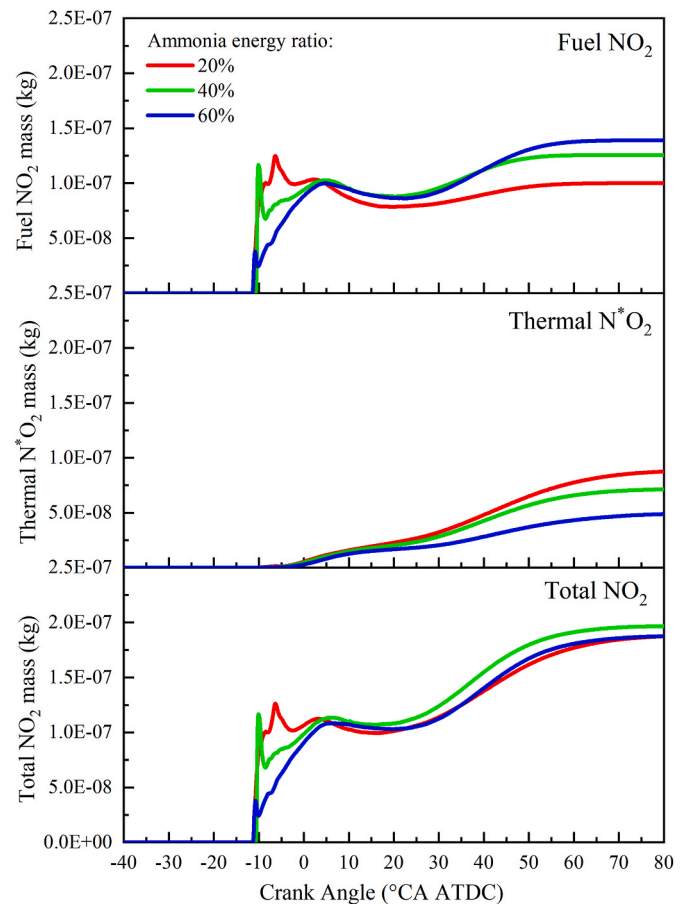


Fig. 21. Calculation results of NO_2 content from different sources at each ammonia energy ratio.

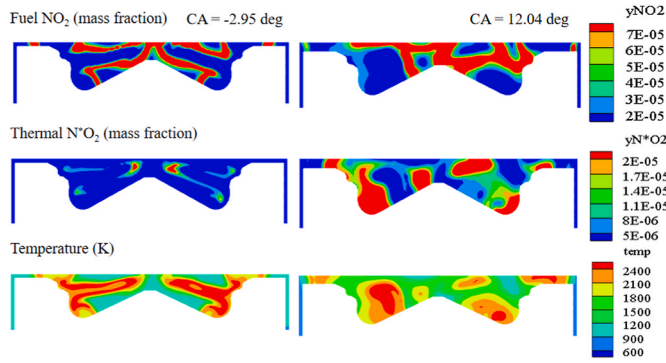
40% NH₃ energy ratio

Fig. 22. Calculation results of NO₂ and temperature distributions with a 40 % ammonia energy ratio.

ratio increased from 20 % to 40 % and 60 %, the fuel NO proportion also increased, which accounted for 35.2 %, 46.1 % and 58.8 %, respectively. Fig. 20 shows the distribution of NO and the temperature field in the cylinder with a 40 % ammonia energy ratio. Notably, the main generation area of fuel NO in the early combustion stage was the diesel injection area; that is, ammonia combustion under diesel ignition generates fuel NO. In the later combustion stage, fuel NO concentrates in the temperature region of about 2000K. The thermal NO generation region markedly coincided with the high-temperature region, which is generated in the temperature region above 2000K, in line with the principle of high-temperature generation.

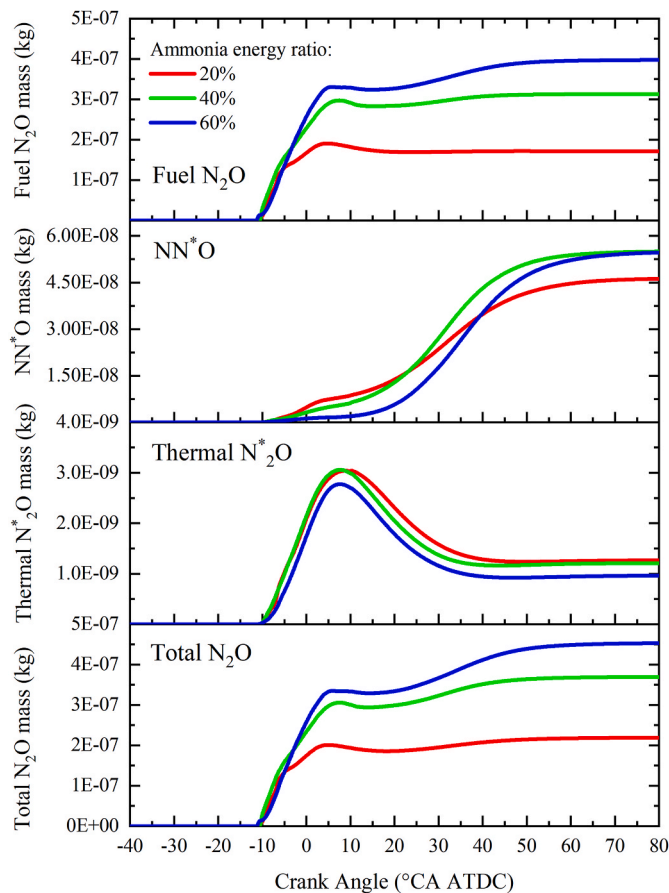


Fig. 23. Calculation results of N₂O content from different sources at each ammonia energy ratio.

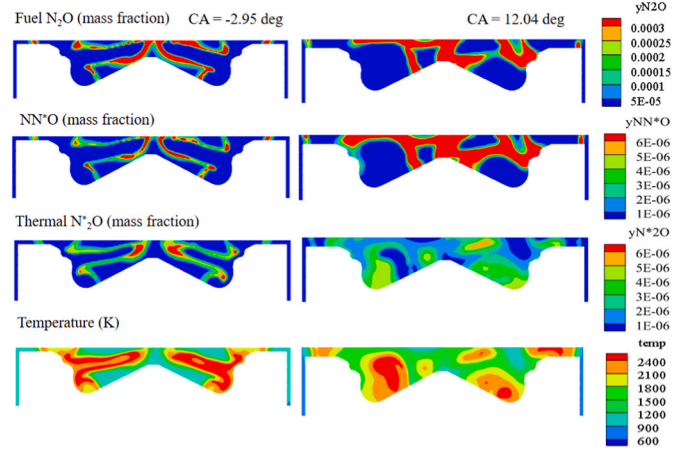
40% NH₃ energy ratio

Fig. 24. Calculation results of N₂O and temperature distributions with a 40 % ammonia energy ratio.

3.4. Analysis of NO₂ emission characteristics

The formation and evolution of NO₂ at 20 %, 40 % and 60 % ammonia energy ratios are shown in Fig. 21. According to the simulation results, the change trend of NO₂ content was mainly observed. At the reaction's early stages, due to NO₂ reduction by ammonia, fuel NO₂ production decreased significantly as the ammonia proportion increased. At the reaction's late stage, as part of NO was oxidized to NO₂, the fuel NO₂ content increased again. Concurrently, thermal N*O₂ formation was slow, which started gradually at the combustion's late stage.

For total NO₂, fuel NO₂ formation in the combustion's early stages was affected by fuel NO₂ content. In the total stable NO₂ content, fuel NO₂ proportion gradually increased as the ammonia energy ratio increased, which were 52.8 %, 63.6 % and 73.6 %, respectively.

The distribution of NO₂ is affected by temperature (Fig. 22). Fuel NO₂ is mainly generated in the temperature region below 1800K. The content of thermal N*O₂ is low and concentrated in the upper part of the combustion chamber. When the temperature is too high, the nitrogen oxide produced is thermal N*O rather than thermal N*O₂.

3.5. Analysis of N₂O emission characteristics

As for N₂O emission (Fig. 23), one of the focuses of the study, the content of fuel N₂O accounts for a very high proportion in the total N₂O content, above 90 %, so the change of total N₂O content is similar to that of fuel N₂O. With the increase of ammonia energy ratio, the content of fuel N₂O increases significantly. Per kg of ammonia burned, 0.008–0.01 kg of fuel N₂O was formed.

Thermal N*₂O is reduced by H in the late reaction stage, and the content is reduced. The formation of NN*O is slow. Their common characteristic is very low content and little effect on total N₂O content.

N₂O content increased as the ammonia energy ratio increased. While decreased combustion temperature promoted N₂O generation, more NO was reduced to generate N₂O under the action of NH. As shown in Fig. 24, compared with NO₂, N₂O was generated in the low-temperature region. At temperatures below 1500K, fuel N₂O is produced in large quantities. Meanwhile, N₂O generation was more significantly affected by temperature. When the combustion temperature decreased due to a increasing ammonia energy ratio, N₂O generation increased significantly. Therefore, reasonable control of in-cylinder temperature is an effective means to reduce N₂O emission.

4. Conclusions

The N in ammonia-diesel mechanism was labeled by isotope labeling method, and the chemical reaction paths of N_2O and NO_x from different sources in ammonia-diesel dual-fuel engine were analyzed. The experiment and simulation of ammonia-diesel dual-fuel engines were conducted, and the spatiotemporal distribution characteristics of N_2O and NO_x from different sources were analyzed. The main conclusions of this study are as follows:

1. Fuel NO was produced in large quantities by ammonia combustion and reacted with NH, O, and N to produce NO_2 , N_2O , and N_2 in the reaction's late stage. Thermal N^*O is formed by oxidation of N^*_2 and reduced to NN^*O and NN^* . When the ammonia energy ratio increased from 20 % to 60 %, the proportion of NO emission in nitrogen oxide emissions decreased from 77.1 % to 71.4 %. The proportion of fuel NO increased from 35.2 % to 58.8 %.
2. According to the experimental data, when the ammonia energy ratio was increased from 20 % to 60 %, total NO_2 emissions reduced by about 60 %, and the NO_2 emission proportion in the nitrogen oxide emission decreased from 15.7 % to 8.5 %. From the trend of the simulation results, the fuel NO_2 proportion in the total NO_2 content positively correlated with the ammonia energy ratios. At a high ammonia energy ratio, fuel NO_2 became the main contributor to NO_2 emissions.
3. For N_2O , the main source of the generation of fuel N_2O are NO and NH, and thermal N^*_2O is mainly generated through the oxidation of N^*_2 . Both of them are consumed by the reaction with H. When the ammonia energy ratio increased from 20 % to 60 %, the increase rate of N_2O emission is approximately doubled, and the N_2O proportion in nitrogen oxide emissions increased from 7.3 % to 20 %. Fuel N_2O accounted for about 90 % of the total N_2O and each kilogram of ammonia combustion generated approximately 0.008–0.01 kg of fuel N_2O .
4. Regarding the spatiotemporal distribution of N_2O and NO_x formation, fuel N_2O and NO_x were mainly generated in the ammonia-rich low-temperature region, and thermal N^*_2O and N^*O_x were mainly generated in the middle- or high-temperature region. High temperature can inhibit the generation of N_2O , so that thermal N^*_2O content was extremely low, and fuel N_2O was concentrated in the low-temperature area. Owing to its high-temperature formation mechanism, thermal nitrogen oxide was generated later than fuel nitrogen oxide and only after the combustion temperature in the cylinder increased.

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Credit authors statement

Concept: WU Binyang. Modeling and Simulation: Feng Yongming. Experiment: Wang Decheng. Formal analysis: Wang Yusong took the main role in results analysis. WU Binyang and Jin Shouying assisted the analysis. Manuscript: The draft was written by Wang Yusong and was thoroughly proof-read by Jin Shouying. Guidance: The team was guided throughout the process by WU Binyang.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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